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ORIGINAL SCIENTIFIC PAPER
Acute Coronary Syndromes

Long-term outcomes of recurrent myocardial infarction: a nationwide Danish study

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Aims

During the past decades outcomes of first-time myocardial infarction (MI) have improved substantially. However, it is unknown if the prognosis following a recurrent MI has also improved similarly.

Methods and results

We conducted a nationwide registry-based study including all patients with first-time recurrent MI in Denmark during 2003 to 2022. Cumulative incidences and standardized risk ratios (sRR) of mortality, hospitalisation for heart failure (HHF), and subsequent recurrent MI were reported along with stratified analyses by age, sex, and HF status at baseline. A total of 24,799 patients with recurrent MI were identified. Between 2003 and 2007 ($n = 7368$) and 2018–2022 ($n = 4928$), their median age decreased from 75 to 73 years. The prevalence of non-cardiovascular comorbidities increased. The use of lipid-lowering treatment at baseline increased (53.7% to 76.6%), as well as procedures performed in relation to recurrent MI (coronary angiogram, 41.2% to 77.4%; percutaneous coronary intervention, 26.8% to 54.0%). 5-year mortality decreased from 54.1% to 37.3% [sRR: 0.78 (0.74–0.82)], 5-year incidence of HHF decreased from 13.6% to 11.7% [sRR: 0.76 (0.68–0.84)], and 5-year incidence of subsequent recurrent MI decreased from 23.4% to 17.7% [sRR: 0.65 (0.52–0.78)]. While mortality and subsequent recurrent MI decreased consistently across subgroups, stratified analyses revealed that the 5-year incidence of HHF increased from 23.9% to 26.2% in patients with previous HF and from 14.3% to 15.9% in males aged ≥75 years.

Conclusion

Mortality has decreased in parallel with intensified pharmacologic and invasive management of patients with recurrent MI. However, there has been little improvement in heart failure hospitalisations, underscoring that directed preventive strategies are needed to mitigate the heart failure risk in patients with recurrent MI.

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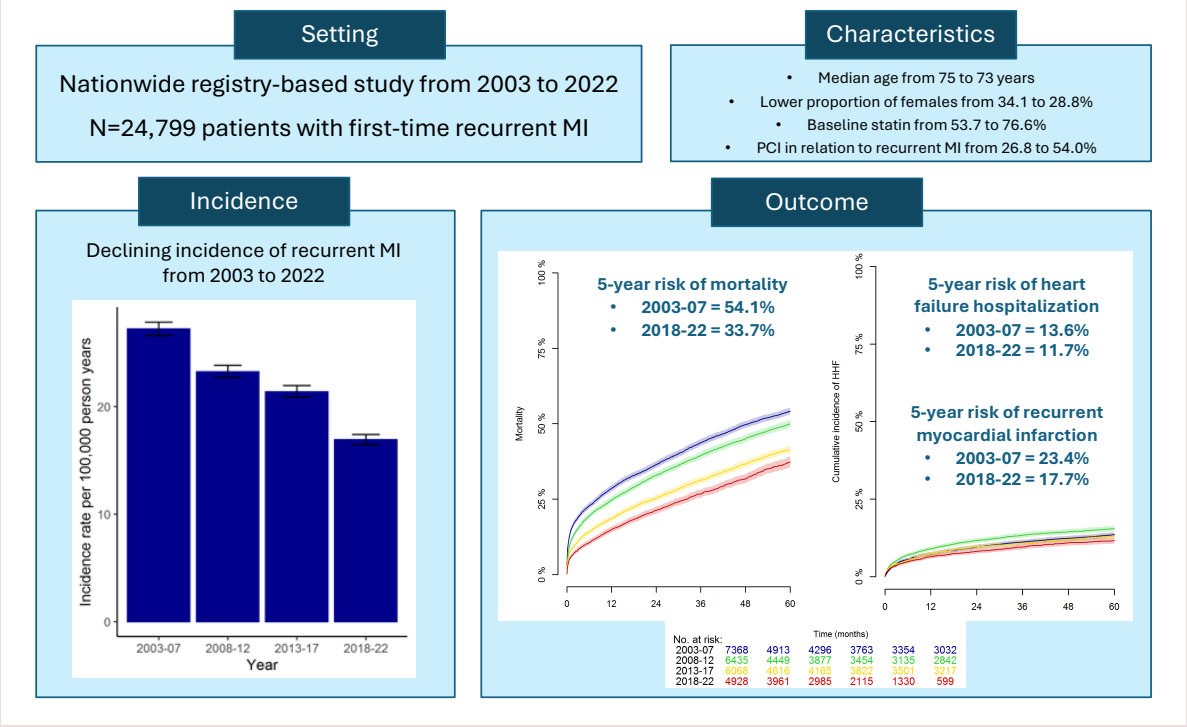
Graphical Abstract



Keywords Recurrent myocardial infarction • Myocardial infarction • Trends • Epidemiology • Outcomes • Registry

Introduction

Implementation of sensitive diagnostics, rapid reperfusion, and guideline-directed medical treatment has correlated with improved in-hospital and long-term outcomes of first-time myocardial infarction (MI) during the past decades across various health care settings.[1–5] The incidence of recurrent MI is ~4–5% at one year and 27% at 10 years among first-time MI patients, with a marked downward trend in recent decades.[6–8] However, smaller cohort studies indicate that patients who suffer a recurrent MI still remain at high risk of subsequent poor outcomes.[9–11] Acquiring adequate data to study these patients is challenged by the low overall incidence of recurrent MI, making large population-based datasets with decades of coverage a prerequisite to sufficiently characterize the population. As such, the epidemiology of recurrent MI in the modern era of cardiology is poorly described.[12]

Comprehensively describing the contemporary recurrent MI population is important to help guide future trials and improve clinical management. Accordingly, we used a nationwide population-based registry, collecting data since 1976, to investigate temporal trends in the incidence, characteristics, and outcomes of recurrent MI between 2003 and 2022.

Methods

Data sources and permissions

We obtained data from the following Danish nationwide registries: The Danish Civil Registration System,[13] the Danish National Patient Registry,[14] the Danish National Prescription Registry,[15] the National Causes of Death Registry,[16] This allowed us to construct an open population-wide longitudinal cohort with accurate information on dates of birth, death, prescriptions (since 1994), and hospital contacts (since 1976). The Danish nationwide registries are of high quality and have been widely used for epidemiological research.[17]

Study population and characteristics

Among the entire Danish population aged >18 years, we identified all patients with a first-time recurrent MI between 2003 and 2022 as follows: All first-time admissions to a hospital in Denmark with a primary discharge diagnosis of MI (ICD-10: I21; ICD-8: 410) between 1976 and 2022 were identified. Patients who were then subsequently admitted to hospital between 2003 and 2022 with a primary diagnosis of MI or recurrent MI (ICD-10: I21, I22) > 28 days after discharge from the initial MI admission were defined as first-time recurrent MI. The diagnosis of recurrent MI has previously been validated in the Danish registries with a positive predictive value (PPV) of 88%.[18] Baseline comorbidities were assessed in the 5 years prior to date of first-time recurrent MI. Baseline pharmacotherapy was assessed in the 180 days prior to date of first-time recurrent MI. See Supplementary material online, *Table S1* for further details.

Outcomes

The outcomes of interest were the incidence of first-time recurrent MI, subsequent all-cause and cardiovascular mortality, incidence of subsequent hospitalisation for heart failure (HHF), and incidence of subsequent recurrent MI. Cardiovascular mortality was defined as a death certificate with a cardiovascular diagnosis [International Classification of Diseases, 10th revision (ICD-10): I00–99] listed as the primary cause of death. HHF was

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Table 1 Characteristics of the study population

	2003–2007 (*n* = 7368)	2008–2012 (*n* = 6435)	2013–2017 (*n* = 6068)	2018–2022 (*n* = 4928)
Age [median (IQR)]	75 [65, 83]	74 [64, 83]	72 [63, 81]	73 [62.8, 80.0]
Males	4859 (65.9)	4355 (67.7)	4263 (70.3)	3509 (71.2)
Years since MI [median (IQR)]	5 [1.2, 11.5]	5.8 [1.6, 12.8]	6.7 [2.1, 13.5]	6.9 [2.2, 14.3]
Previous CABG	609 (8.3)	852 (13.2)	911 (15.0)	706 (14.3)
Previous PCI	1692 (23.0)	2766 (43.0)	3603 (59.4)	3448 (70.0)
Baseline comorbidities				
Atrial fibrillation	1008 (13.7)	979 (15.2)	933 (15.4)	662 (13.4)
CKD	401 (5.4)	488 (7.6)	551 (9.1)	433 (8.8)
COPD	774 (10.5)	769 (12.0)	667 (11.0)	414 (8.4)
Heart failure	1784 (24.2)	1518 (23.6)	1267 (20.9)	842 (17.1)
Peripheral artery disease	948 (12.9)	923 (14.3)	817 (13.5)	643 (13.0)
Cancer	559 (7.6)	534 (8.3)	581 (9.6)	490 (9.9)
Stroke	744 (10.1)	559 (8.7)	371 (6.1)	274 (5.6)
Hypertension	3994 (54.2)	3890 (60.5)	3415 (56.3)	2660 (54.0)
Diabetes	1387 (18.8)	1444 (22.4)	1514 (25.0)	1323 (26.8)
Baseline pharmacotherapy				
ADP-receptor inhibitor	1564 (21.2)	1825 (28.4)	1938 (31.9)	1650 (33.5)
Aspirin	5897 (80.0)	5279 (82.0)	4564 (75.2)	3362 (68.2)
Lipid-lowering drug	3960 (53.7)	4797 (74.5)	4573 (75.4)	3775 (76.6)
Oral anticoagulant	554 (7.5)	617 (9.6)	848 (14.0)	847 (17.2)
Beta blocker	4523 (61.4)	4215 (65.5)	3970 (65.4)	2973 (60.3)
ACEi	2888 (39.2)	2827 (43.9)	2193 (36.1)	1500 (30.4)
ARB	1004 (13.6)	1273 (19.8)	1408 (23.2)	1411 (28.6)
MRA	869 (11.8)	713 (11.1)	564 (9.3)	453 (9.2)
Procedures performed in relation to recurrent MI admission				
Coronary angiogram	3038 (41.2)	3656 (56.8)	4203 (69.3)	3813 (77.4)
PCI	1975 (26.8)	2249 (34.9)	2788 (45.9)	2659 (54.0)
CABG	526 (7.1)	407 (6.3)	383 (6.3)	284 (5.8)

ACEi, angiotensin-converting enzyme inhibitor; ADP, adenosine diphosphate; ARB, angiotensin receptor blocker; CABG, coronary artery bypass graft; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; IQR, interquartile range; MI, myocardial infarction; MRA, mineralocorticoid receptor antagonist; PCI, percutaneous coronary intervention.

defined as an inpatient hospitalisation with a primary discharge diagnosis of heart failure (ICD-10: I132, I50, I110). A primary diagnosis of heart failure in the Danish National Patient Registry has previously been validated with a PPV of 88%.[19]

Statistical analyses

The study period was divided into 5-year intervals (2003–2007, 2008–2012, 2013–2017 and 2018–2022). Categorical variables were reported as frequencies with percentages for and continuous variables as medians with interquartile ranges.

Incidence rates per 100 000 person years of first-time recurrent MI with 95% confidence intervals were calculated for each 5-year calendar period. For each 5-year period, the entire Danish population was followed from the start of the period (1 January 2003, 1 January 2008, etc.) until death, first-time recurrent MI, or the end of the 5-year period (31 December 2007, 31 December 2012, etc.), whichever came first.

Cumulative incidences of mortality, HHF, and subsequent recurrent MI were calculated using the Kaplan-Meier method for mortality and the Aalen-Johansen method for cardiovascular mortality, HHF and subsequent recurrent MI. For the outcomes of HHF and subsequent recurrent MI, death without outcome was considered a competing risk. We fitted cause-specific Cox regression models adjusted for age (dichotomized), sex, years since first MI (dichotomized), baseline comorbidities, baseline pharmacotherapy, and previous coronary revascularization for the outcomes of interest and the competing risk of death. Based on these models, we calculated standardized 5-year risk ratios (sRR) comparing each calendar year period to the reference period.[20] Time zero of follow-up was set at date of first-time recurrent MI. Patients were followed until date of outcome, date of death, or 31 December 2023, whichever came first.

All analyses were repeated in strata of age (<75, ≥75 years) and sex. Outcome analyses were repeated in groups defined by previous heart failure vs. no previous heart failure and by ST-elevation myocardial infarction (STEMI) vs. non-STEMI or unspecified MI. The level of statistical significance was set at 5%. All analyses were performed using R version 4.2.1.

Ethics

The Danish Data Agency approved this project (project number P-2019-191). In Denmark, registry-based studies do not require ethical approval.

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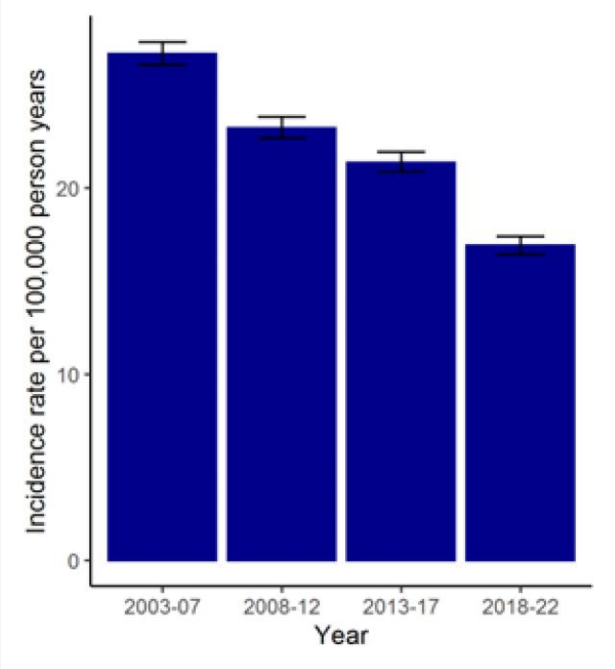

Figure 1 Incidence rate (per 100 000 person years, with 95% confidence intervals) of first recurrent myocardial infarction in the general population between 2003 and 2022.

Results

Between 2003 and 2022, a total of 24 799 patients with a first-time recurrent MI were identified. During the study period, the median age increased from 73 to 75 years. The proportion of males increased from 65.9% to 71.2%. The proportion of patients with previous heart failure and previous stroke decreased from 24.2% to 17.1% and from 10.1% to 5.6%, respectively, whereas the proportion of non-cardiovascular comorbidities such as diabetes, cancer, and chronic kidney disease (CKD) increased (*Table 1*).

The use of lipid-lowering treatment (53.7% to 76.6%) and ADP-receptor inhibitors (21.2% to 33.5%) at baseline increased, while an increase in the use of oral anticoagulants was offset by a decrease in the use of aspirin (*Table 1*). The proportion of patients receiving a coronary angiogram in relation to their recurrent MI increased from 41.2% to 77.4%, while the proportion of patients receiving a percutaneous coronary intervention (PCI) increased 26.8% to 54.0%. Older patients and patients with previous HF had more comorbidities, less use of evidence-based pharmaceutical treatment at baseline, and lower use of coronary procedures in relation to the recurrent MI (see Supplementary material online, *Tables S2* and *S3*).

Incidence of first-time recurrent MI

The overall incidence of first-time recurrent MI in the general population decreased from 27.4 (95% CI: 26.6–27.9) to 16.9 (95% CI: 16.5–17.4) per 100 000 person years between 2003 and 2022 (*Figure 1*). This decrease was observed across subgroups of age and sex (see Supplementary material online, *Figure S1*). Males aged ≥75 years had

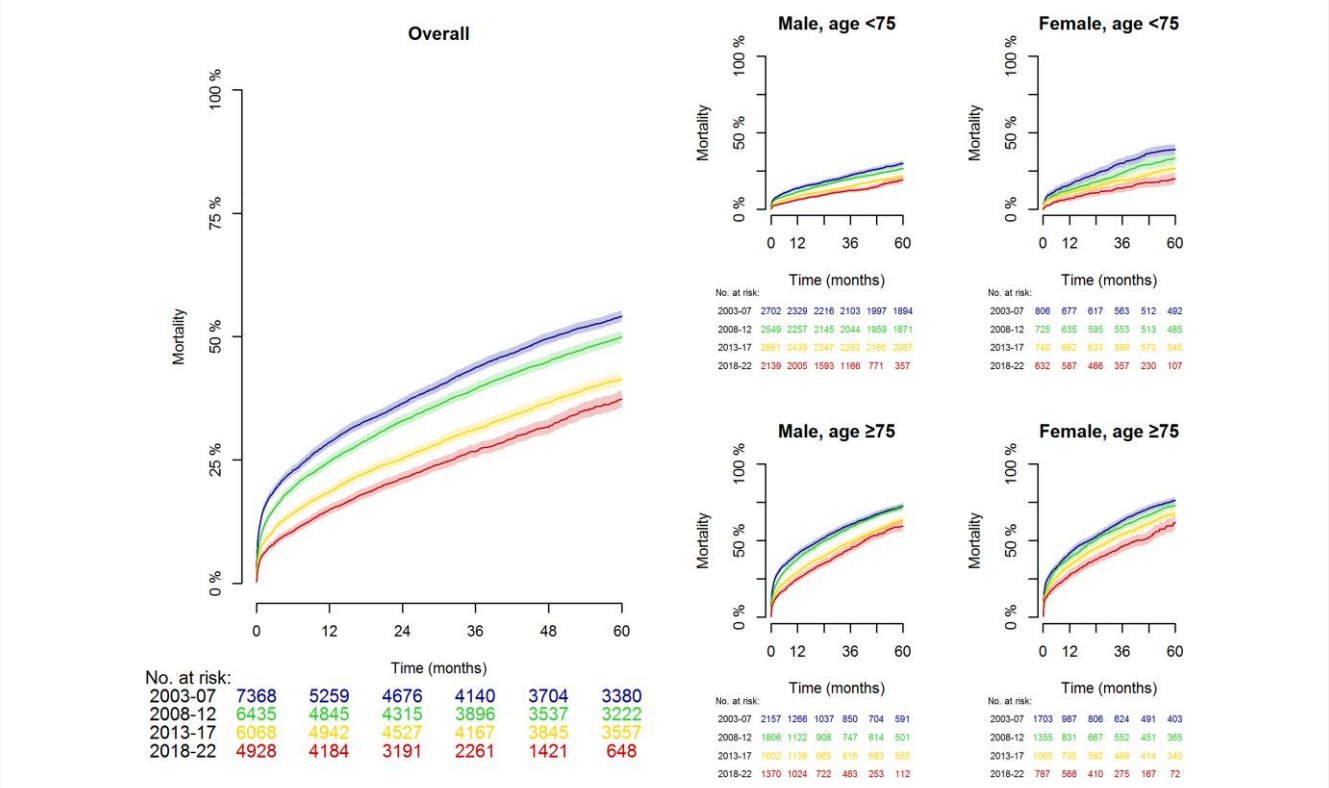

Figure 2 Cumulative incidence of mortality in patients with a first recurrent myocardial infarction (with 95% confidence intervals) between 2003 and 2022, overall and in subgroups of age and sex.

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the highest incidence of recurrent MI but also had the highest absolute decrease; from 302.0 (95% CI: 289.4–315.0) to 123.8 (95% CI: 117.3–130.5) per 100 000 person years.

Outcomes following first-time recurrent MI

The cumulative incidence of mortality declined significantly between 2003 and 2022. Overall, 5-year mortality declined from 54.1% (95% CI: 53.0–55.3) to 37.3% (35.6–39.1). This decline was consistent across subgroups of age and sex (*Figure 2*), e.g. 5-year mortality dropped from 39.0% (95% CI: 35.6–42.3) to 19.9% (95% CI: 16.0–23.8) in females aged <75 years. A similar pattern was noted for subsequent MI: overall, the 5-year cumulative incidence was 23.4% (95% CI: 22.5–24.4) in 2003–07 vs. 17.7% (95% CI: 16.4–18.9) in 2018–22 (see Supplementary material online, *Figure S2*), which was consistent in subgroups (see Supplementary material online, *Figure S3*) and in adjusted analyses (see Supplementary material online, *Figure S4*).

A less notable decline was observed in the cumulative incidence of HHF: overall, the 5-year cumulative incidence declined from 13.6% (95% CI: 12.8–14.4) to 11.7% (95% CI: 10.7–12.8). Subgroup analyses revealed a lack of improvement in older patients; during the study period, the 5-year cumulative incidence of HHF in males aged <75 years declined from 12.5% (95% CI: 11.3–13.8) to 9.4% (95% CI: 8.0–10.9) while it increased from 14.3% (95% CI: 12.8–15.8) to 15.9% (95% CI: 13.6–18.3) in males aged ≥75 years (*Figure 3*).

Additional analyses

The 5-year mortality declined similarly in patients with vs. without previous HF, albeit with higher mortality in those with previous HF: 59.7% (95% CI: 57.0–62.4) vs. 33.2% (95% CI: 31.3–35.1) in 2018–22 (*Figure 4*).

In those without previous HF, the 5-year cumulative incidence of HHF declined from 10.3% (95% CI: 9.5–11.1) to 8.7% (95% CI: 7.7–9.8) during the study period, whereas it increased from 23.9% (95% CI: 21.9–25.9) to 26.2% (95% CI: 23.0–29.4) in those with previous HF (*Figure 4*).

SRRs accounting for changes in patient characteristics during the study period showed similar results to the main analyses (*Figure 5*). In the Cox regression models underlying these standardized risk ratios, age was found to be the most significant predictor of mortality, while previous HF was the most significant predictor of HHF, and CKD was the most significant predictor of subsequent recurrent MI (see Supplementary material online, *Figures S5–S7*).

An analysis stratified by type of MI showed similar results to the main analysis (see Supplementary material online, *Figure S8*). The 5-year mortality declined from 43.0% to 27.7% in patients who presented with STEMI, while 5-year cumulative incidence of HHF went from 13.4% to 11.1%. Patients with a diagnosis code of STEMI were younger and had more coronary procedures compared with the overall study population (see Supplementary material online, *Table S4*).

Cardiovascular mortality at 5 years declined from 35.2% in 2003–2007 to 14.0% in 2018–2022 (see Supplementary material online, *Figure S9*).

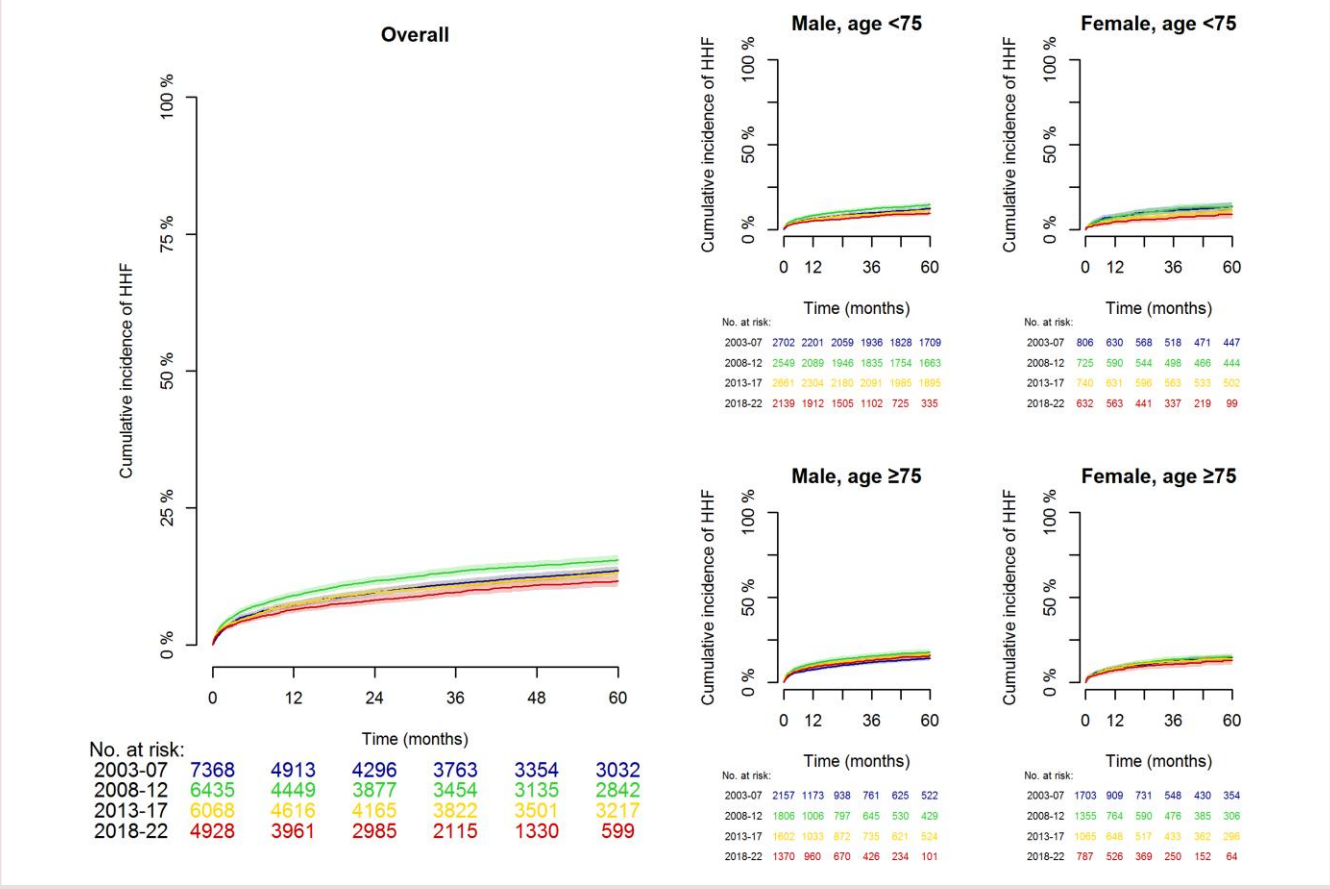

Figure 3 Cumulative incidence of hospitalisation of heart failure in patients with a first recurrent myocardial infarction (with 95% confidence intervals) between 2003 and 2022, overall and in subgroups of age and sex.

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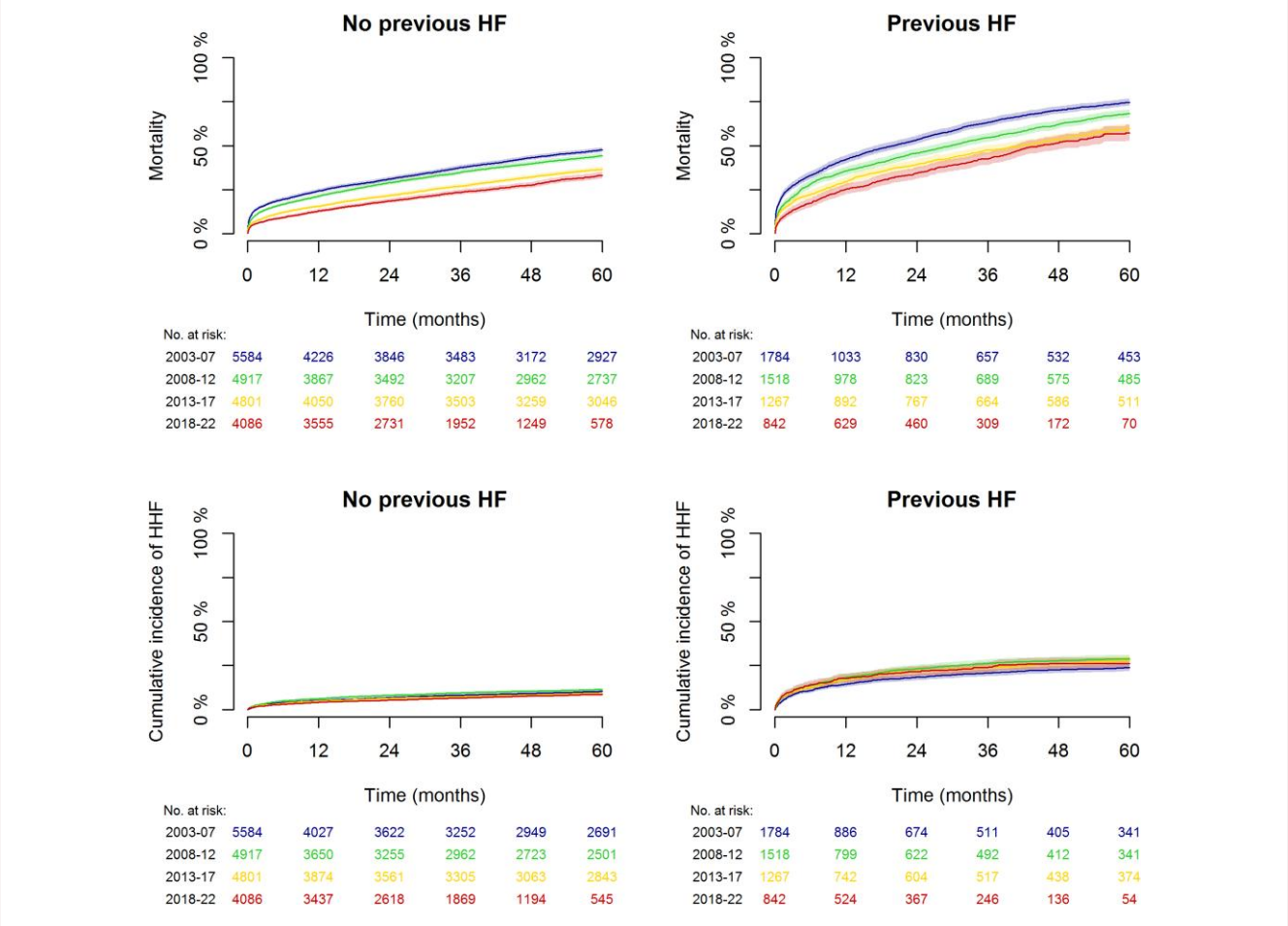

Figure 4 Cumulative incidence of mortality and hospitalisation of heart failure in patients with a first recurrent myocardial infarction (with 95% confidence intervals) between 2003 and 2022, according to heart failure status at baseline.

Discussion

Our data from a large nationwide cohort showed a considerable decline in the incidence of recurrent MI during the past two decades. The long-term risk of mortality and subsequent recurrent MI has improved substantially over time. In contrast, the long-term risk of HHF following a recurrent MI did not change markedly.

Few studies have analyzed the epidemiology of recurrent MI in the era of modern cardiology, due to the relatively low incidence and the comprehensive datasets required.[12] Not surprisingly, we showed a declining incidence of recurrent MI, as it has been well demonstrated that both incident first-time MI as well as cardiovascular events following a first-time MI have declined drastically in recent decades.[4,8,21,22] However, it was unclear whether this improvement also pertained specifically to patients with recurrent MI. The rates of CAG and PCI during initial recurrent MI hospitalisation increased over time, and were in 2018–2022 similar to rates reported in all-comers real-world MI populations.[1,23–25] Smaller studies have shown that recurrent MI is associated with a worse prognosis compared with first-time MI.[10,11,26] However, a study from the pre-rapid reperfusion era showed improvements in the long-term risks of mortality and cardiovascular events in both first-time and recurrent MI.[27] Interestingly, it showed that the gap in prognostic impact between first-time and recurrent MI was diminishing between 1985 and 2002. Our data shows that this gap persists, which could partly be explained by the older age and higher comorbidity burden of patients with recurrent MI. Of particular note, the gap may be partly driven by the risk of HHF, which barely improved over two decades. However, the modest improvement in risk of HHF over time may in fact be partly explained by the improved survival. That is, patients in previous periods with a high risk of HF or pre-existing HF were more likely to die after their recurrent MI, but may now survive long enough to develop HHF.

Favourable trends in incidence and outcomes of MI have been linked with various factors, namely improvements in risk factors for ischaemic heart disease such as smoking, elevated lipid levels, and elevated blood pressure,[22,28] optimized medical secondary prevention and rapid reperfusion,[1,2] and improved sensitivity of diagnostic methods.[29] These circumstances likely also contributed to the observed improved outcomes following recurrent MI. However, our data underscores that patients with recurrent MI represent a distinct patient population compared with first-time or all-comers MI. In comparison with first-time MI patients, patients in our study were considerably older (median age 73 years in 2018–2022 vs. 67 years for first-time MI) and had more comorbidities, such as CKD, diabetes, or previous heart failure.[6] In particular, patients with recurrent MI represent a population with complex ischaemic heart disease, among whom 14.3% had previous CABG surgery and 70% had a previous PCI, in the most recent period. These factors likely contribute to explaining the high long-term risks of adverse

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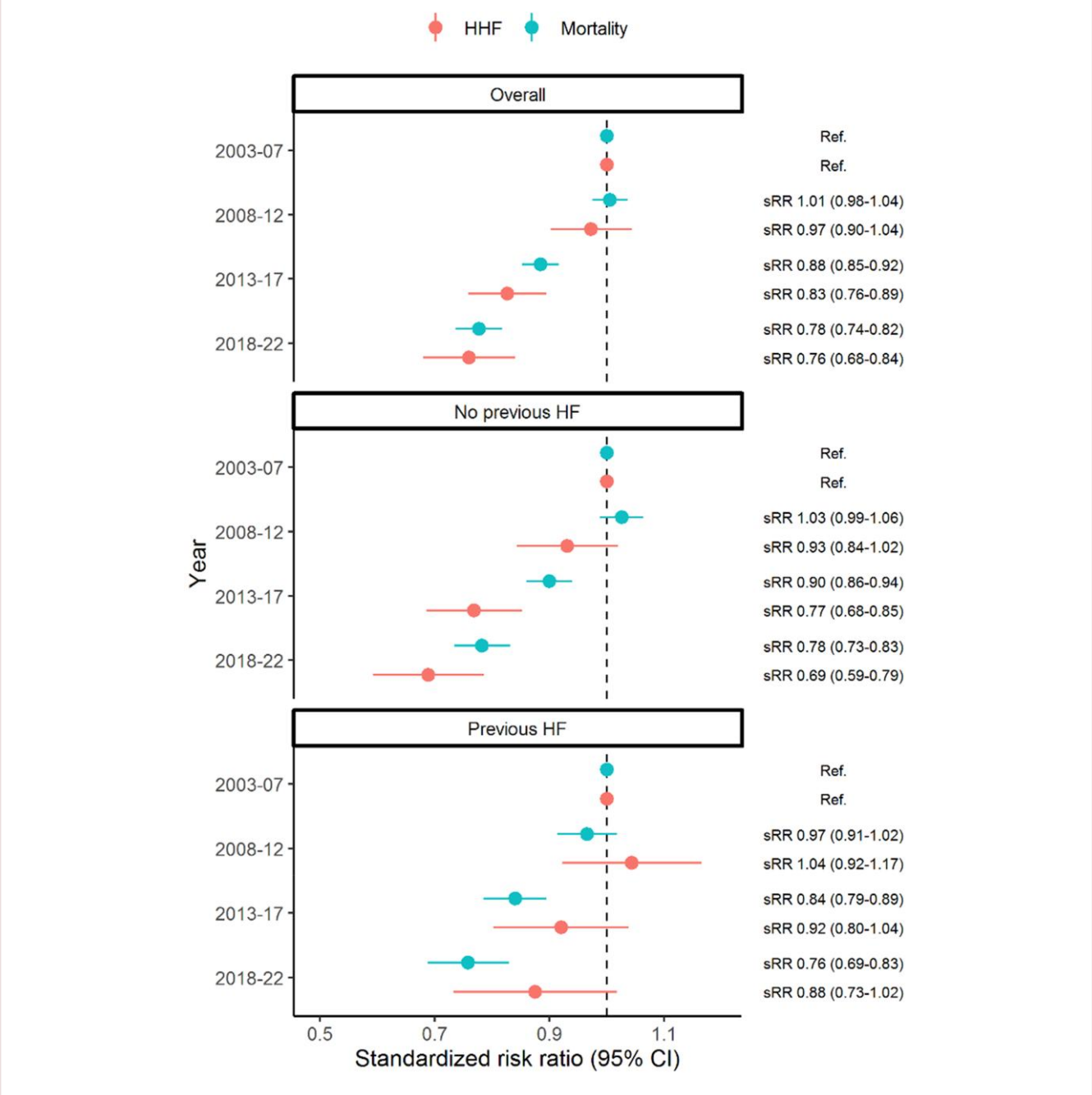

Figure 5 Standardized risk ratios with 95% confidence intervals for mortality and hospitalisation for heart failure in patients with a first recurrent myocardial infarction between 2003 and 2022 with the earliest calendar year period serving as reference, overall and in subgroups according to heart failure status at baseline. Standardized for age, sex, time since first myocardial infarction, baseline comorbidities, previous revascularization, and baseline pharmacotherapy.

outcomes that remain in our study population despite the observed temporal improvements. This was particularly pronounced among patients with pre-existing HF, which may have contributed to their poor outcomes. The 5-year risks of mortality and HHF has been reported to be as low as 12.5% and 2.4%, respectively in contemporary stable post first-time MI patients,[3] which stands in contrast to the 37.3% and 11.7% 5-year risks observed in the present study. While previous data have indicated that cardiovascular mortality has improved in well-treated first-time MI patients to such a degree that non-cardiovascular outcomes are becoming dominant,[30–32] our data underline that the residual cardiovascular risk in patients with recurrent MI still remains substantial. Recent trials of potential new pharmaceutical treatments for MI failed to meet their primary endpoints in the younger, healthier trials populations,[33,34] but may yet warrant testing in the higher-risk population of patients with recurrent MI to prevent heart failure and address the residual risk.[35,36]

Guidelines distinguish older patients at an age-cutoff of 75 years,[37] and have previously recommended applying the same diagnostic,

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interventional, and secondary preventive strategies for older patients as for younger patients.[38,39] However, recent guidelines and scientific society statements have acknowledged the complexity in treatment of older patients, as they are often not represented in clinical trials due to comorbidity, complexity, etc.[40,41] These concerns likely apply to patients with a recurrent MI in particular. Indeed, our age-stratified analyses showed that the incidence of recurrent MI is manyfold higher among people aged ≥75 years. Also, in this age group, despite the improvement in survival over time, there was a lower use of lipid-lowering treatment at baseline, a lower use of coronary procedures in relation to recurrent MI, and mortality remained high compared with their younger counterparts. Future trials are needed to provide more evidence-based treatment strategies for this complex patient population.

Limitations

The main strength of our study was the completeness our registries, which enabled to identify all hospitalisations for MI since 1976. This also allowed for minimal loss to follow-up and selection bias. However, a few limitations should be considered when interpreting our results. Importantly, while our registry-based definition of recurrent MI has previously been validated with a high PPV,[18] we cannot rule out misclassification. This means that patients with type 2 MI, myocardial injury, chest pain without MI, etc. may wrongly have been classified as recurrent MI, biasing our findings. Likewise, many patients in our registry were coded with a diagnosis of DI21 (MI, not further specified). This resulted in a low sensitivity for type of MI, despite a good PPV for STEMI (96%). As such, the analyses stratified by type of MI, should be interpreted with caution. Furthermore, our study design did not allow for mediation analysis, to reveal which factors could have driven our results. In adjusted analyses accounting for changes in patient and demographic characteristics, they appeared not to influence the direction nor magnitude of the results. However, highly important variables such as HbA1c, lipid status, smoking, body mass index, ECGs, troponin levels, angiographic findings, discharge prescriptions, and echocardiographic findings were unavailable in our registries. This lack of granularity likely resulted in residual confounding in the adjusted analyses.

Conclusions

During the past two decades, incidence of recurrent MI has declined significantly. Long-term outcomes following recurrent MI have also improved in parallel with guideline-based management during this period, especially with regards to mortality and subsequent MI. However, patients with recurrent MI are a complex population with a high degree of comorbidity. Particular emphasis on preventing HF following recurrent MI is warranted.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Author contributions

Daniel Mølager Christensen (Conceptualization, Formal Analysis, Investigation, Methodology, Visualization, Writing—original draft), Deewa Zahir (Methodology, Writing—review & editing), Caroline Garred (Methodology, Writing—review & editing), Nina Nouhravesh (Methodology, Writing—review & editing), Mariam Elmegaard (Methodology, Writing—review & editing), Morten Malmborg (Methodology, Writing—review & editing), Caroline Sindet-Pedersen (Methodology, Writing—review & editing), Abdullahi Mohamed (Methodology, Writing—review & editing), Mohamad El-Chouli (Methodology, Writing—review & editing), Emil Fosbøl (Investigation, Methodology, Writing—review & editing), Charlotte Andersson (Methodology, Writing—review & editing), Marc A. Pfeffer (Conceptualization, Methodology, Writing—review & editing), Lars Køber (Investigation, Methodology, Writing—review & editing) and Morten Schou (Conceptualization, Data curation, Investigation, Methodology, Supervision, Writing—review & editing).

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Data availability

The data analyzed in this study were accessed through Statistics Denmark. This data cannot be made publicly available as access must be granted to institutions by the Danish Data Protection Agency and The Danish Health Data Authority.

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